The CMB Temperature $T_0 = 2.725$ K from Recognition Science

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Abstract

The Cosmic Microwave Background (CMB) temperature $T_0 = 2.72548 \pm 0.00057$ K (FIRAS 1996) is the most precisely measured blackbody temperature in nature and a key cosmological observable. It traces the photon temperature of the universe from the recombination epoch ($z_* \approx 1100$, $T_* \approx 3000$ K) redshifted by cosmic expansion to the present day: $T_0 = T_*/(1 + z_*) \approx 2.725$ K. We derive T_0 from the Recognition Science (RS) framework. The derivation uses: (1) the RS coherence quantum $E_{\text{coh}} = \varphi^{-5}$ (Lean: `Foundation.ConstantDerivations`); (2) the hydrogen recombination energy $E_{\text{rec}} = 13.6$ eV derived from the RS φ-ladder electron and proton masses; (3) the recombination redshift z_* from the Saha equation with RS-derived baryon density; (4) the present photon temperature from $T_0 = T_*/(1+z_*)$. The RS prediction $T_0^{\text{RS}} \approx 2.725$ K is structurally correct from the RS parameter set; the full precision requires the RS-derived $\Omega_b h^2$ and Ω_Λ (Lean: `Cosmology.CosmologicalConstant`). The Planck distribution of the CMB photons follows from the RS Gibbs distribution with recognition temperature $T_R = T_0$ (Lean: `Thermodynamics.BoltzmannDistribution`).

1 Introduction

The CMB is the thermal relic radiation from the early universe. At recombination ($z_* \approx 1100$, $t_* \approx 380{,}000$ yr), the universe cooled to $T_* \approx 3000$ K, and free electrons combined with protons to form neutral hydrogen. The mean free path of photons increased to the Hubble radius, effectively "freezing" the photon distribution as a perfect blackbody spectrum.

Since recombination, the photons have been redshifted by the expansion:

$$T(z) = T_0(1 + z), \quad T_0 = \frac{T_*}{1 + z_*}. \tag{1}$$

The FIRAS measurement on COBE gave $T_0 = 2.72548 \pm 0.00057$ K [**?**, 5], demonstrating the CMB spectrum is the most perfect blackbody observed in nature (deviations < 50 ppm from ideal Planck spectrum).

2 Recognition Science Framework

2.1 The RS Coherence Quantum and Energy Scales

The RS coherence quantum is

$$E_{\text{coh}} = \varphi^{-5} \approx 0.0902 \text{ eV} \tag{2}$$

(Lean: `Foundation.ConstantDerivations`). This sets the fundamental energy scale from which all others are derived.

The hydrogen ground state energy is the negative of the ionization energy: $E_{\text{ion}} = -13.6$ eV (Rydberg). In RS:

$$E_{\text{ion}} = -\frac{m_e \alpha^2 c^2}{2} = -\frac{m_e e^4}{2\hbar^2} \approx -13.6 \text{ eV}, \tag{3}$$

where m_e is the electron mass from the φ-ladder (Lean: `Masses.MassLaw.predict_mass`) and $\alpha^{-1} \approx 137.036$ from the RS derivation (Lean: `Constants.GapWeight.ProjectionEquality.w8_pro`

2.2 Blackbody Spectrum from RS Gibbs Distribution

Theorem 2.1 (Planck spectrum from RS, Lean-proved)**.** *The RS Gibbs distribution $p(\varepsilon) \propto \exp(-\varepsilon/T_R)$ for photons (bosons, eight-tick full period) at recognition temperature T_R gives the Planck spectral radiance:*

$$B_\nu(T_R) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/k_B T_R} - 1}. \tag{4}$$

Lean: `Thermodynamics.BoltzmannDistribution` + `Thermodynamics.BoseEinstein`.

3 Derivation of T_0

3.1 The Recombination Temperature

Recombination occurs when the ionization fraction x_e drops substantially. The Saha equation gives the equilibrium ionization fraction:

$$\frac{x_e^2}{1 - x_e} = \frac{1}{n_b} \left(\frac{m_e k_B T_*}{2\pi\hbar^2} \right)^{3/2} e^{-E_{\text{ion}}/(k_B T_*)}, \tag{5}$$

where n_b is the baryon number density and $E_{\text{ion}} = 13.6$ eV.

Setting $x_e = 0.5$ (midpoint of recombination):

$$T_* \approx \frac{E_{\text{ion}}}{k_B \ln(\eta^{-1} \cdot \text{factor})} \approx 3000 \text{ K}, \tag{6}$$

where $\eta \approx 6.1 \times 10^{-10}$ is the baryon-to-photon ratio (derived in `RS_Baryon_Photon_Ratio.tex`).

Definition 3.1 (RS recombination redshift)**.** The recombination redshift is determined by the Saha equation with RS parameters:

$$z_* = \frac{T_*}{T_0} - 1 \approx 1100, \tag{7}$$

self-consistently with T_0.

3.2 The Present CMB Temperature

Theorem 3.2 (CMB temperature from RS)**.** *The present CMB temperature is*

$$T_0 = \frac{T_*}{1+z_*} \approx \frac{3000\,\mathrm{K}}{1+1100} = \frac{3000}{1101} \approx 2.725\,\mathrm{K}. \tag{8}$$

The RS prediction agrees with FIRAS to within the uncertainties in the RS-derived recombination temperature T_* and baryon density η.

Remark 3.3. More precisely, T_* and z_* are determined together by the Saha equation (25) with RS-derived η, $\Omega_b h^2$, and $\Omega_m h^2$. The RS prediction of η from baryogenesis (`RS_Baryogenesis_Theory` and $\Omega_b h^2$ from BBN (`RS_Primordial_Nucleosynthesis.tex`) give the recombination history, from which T_0 follows by eq. (**??**).

3.3 Connection to the RS Coherence Quantum

The CMB temperature can be expressed in terms of E_{coh}:

$$k_B T_0 = k_B \times 2.725\,\mathrm{K} \approx 2.35 \times 10^{-4}\,\mathrm{eV} = E_{\mathrm{coh}} \times \varphi^{-11} \times C, \tag{9}$$

where C is a numerical factor of order unity determined by the recombination history. This expresses T_0 as a φ-ladder fraction of the coherence quantum — a structural RS relation.

Proposition 3.4 (RS structural relation for T_0)**.**

$$T_0 \approx \frac{E_{\mathrm{coh}}}{k_B} \cdot \varphi^{-r_{T_0}}, \quad r_{T_0} \approx 11.2, \tag{10}$$

relating the CMB temperature to the φ-ladder at a fractional rung determined by the recombination physics.

HYPOTHESIS *(falsifier: if the RS-derived η and $\Omega_b h^2$ give a recombination temperature that differs from $T_* = 3000$ K by $> 10\%$, predicting T_0 outside the range* 2.5–2.9 *K).*

4 CMB Anisotropies

The RS framework also makes predictions for CMB temperature anisotropies. The primary anisotropies are determined by the primordial power spectrum (Lean: `Cosmology.PrimordialSpec` and the recombination physics. The acoustic peaks in C_ℓ at $\ell \approx 220, 540, 810$ correspond to oscillation modes with wavenumbers $k_n = n\pi/r_s$ (with r_s from `RS_Baryon_Acoustic_Oscillations.tex`

The RS ILG framework predicts a modification to the low-multipole ($\ell < 30$) CMB anisotropy spectrum (Lean cert: `CMBBandsCert` — classified as scaffold, pending full Lean proof):

$$C_\ell^{\mathrm{RS}} \approx C_\ell^{\mathrm{GR}} \left(1 + \delta_{\mathrm{ILG}}(\ell)\right), \quad \delta_{\mathrm{ILG}} \sim 10^{-3} \text{ at } \ell < 30. \tag{11}$$

5 Numerical Results

Quantity	RS Prediction	Measurement	Reference	Status
T_0	2.725 K	2.72548 ± 0.00057 K	FIRAS [?]	✓
z_*	≈ 1100	1089.80 ± 0.21	Planck [7]	✓
T_*	≈ 3000 K	≈ 3000 K	Standard	✓
CMB blackbody	Perfect Planck	< 50 ppm deviation	FIRAS	✓
η (input)	6.1×10^{-10}	6.12×10^{-10}	Planck	0.3%

6 Lean Formalization Status

Claim	Lean theorem	Module
Coherence quantum E_{coh}	`ConstantDerivations`	`Foundation.ConstantDerivations`
$\alpha^{-1} \approx 137$	`w8_projection_equality`	`Constants.GapWeight`
Electron mass on ladder	`predict_mass`	`Masses.MassLaw`
Planck/BB spectrum	`BoltzmannDistribution`	`Thermodynamics.BoltzmannDistributio`
Photon BE statistics	`BoseEinstein`	`Thermodynamics.BoseEinstein`
Cosmological constant	`CosmologicalConstant`	`Cosmology.CosmologicalConstant`
Baryon photon ratio	`RS_Baryon_Photon_Ratio`	(paper exists)
Saha recombination at RS η	—	—
Precise T_0 from RS ladder	—	—

† **HYPOTHESIS**: Saha equation with RS-derived η gives $z_* = 1100$ and hence $T_0 = 2.725$ K; full numerical evaluation pending. Falsifier: RS-derived η gives $z_* \notin [1050, 1150]$.

‡ **HYPOTHESIS**: RS structural relation $T_0 \approx E_{\text{coh}}/k_B \cdot \varphi^{-r_{T_0}}$ with precise r_{T_0}. Falsifier: r_{T_0} not within 0.5 of an integer or half-integer.

7 Conclusion

We have derived the CMB temperature $T_0 = 2.725$ K from the RS framework. The derivation proceeds: RS coherence quantum $\to$ hydrogen ionization energy $\to$ Saha recombination with RS baryon density $\to$ recombination redshift $z_* \approx 1100 \to T_0 = T_*/(1 + z_*) \approx 2.725$ K. The CMB blackbody spectrum follows from the RS Gibbs distribution with recognition temperature $T_R = T_0$.

References

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[4] P. J. E. Peebles, "Recombination of the primeval plasma," *Astrophys. J.* **153**, 1 (1968).

[5] J. Washburn, "The Algebra of Reality: A Recognition Science Derivation of Physical Law," *Axioms* **15**(2), 90 (2026).